

Heart Disease — Inference Demo

Loads the persisted `models/model.pkl` and runs predictions for single + batch patient inputs.

```
In [1]: import sys
        from pathlib import Path
        import joblib
        import pandas as pd

        ROOT = Path.cwd().parent if Path.cwd().name == 'notebooks' else Path.cwd()
        sys.path.insert(0, str(ROOT))
```

1. Load the persisted pipeline

```
In [2]: model = joblib.load(ROOT / 'models' / 'model.pkl')
        print('Pipeline steps:', [s for s, _ in model.steps])
        print('Final estimator:', type(model.named_steps['model']).__name__)
```

```
Pipeline steps: ['preprocessor', 'model']
Final estimator: LogisticRegression
```

2. Single-patient prediction

```
In [3]: sample = {
        'age': 54, 'sex': 1, 'cp': 0, 'trestbps': 130, 'chol': 250,
        'fbs': 0, 'restecg': 1, 'thalach': 150, 'exang': 0,
        'oldpeak': 1.0, 'slope': 1, 'ca': 0, 'thal': 2,
        }
        df_one = pd.DataFrame([sample])
        proba = model.predict_proba(df_one)[0, 1]
        label = int(model.predict(df_one)[0])
        print(f'P(disease) = {proba:.3f} | predicted label = {label}')
```

```
P(disease) = 0.097 | predicted label = 0
```

3. Batch prediction

```
In [4]: batch = pd.DataFrame([
        sample,
        {'**sample', 'age': 35, 'chol': 180, 'thalach': 190, 'oldpeak': 0.0},
        {'**sample', 'age': 70, 'chol': 320, 'oldpeak': 4.5, 'ca': 3},
        ], index=['baseline', 'low_risk', 'high_risk'])
        batch['p_disease'] = model.predict_proba(batch)[:, 1].round(3)
        batch['pred'] = model.predict(batch)
        batch[['age', 'chol', 'thalach', 'oldpeak', 'ca', 'p_disease', 'pred']]
```

Out[4]:

	age	chol	thalach	oldpeak	ca	p_disease	pred
baseline	54	250	150	1.0	0	0.097	0
low_risk	35	180	190	0.0	0	0.039	0
high_risk	70	320	150	4.5	3	0.657	1